

EEE3095/6S: Practical 2

Embedded ARM Assembly Programming

August 13, 2025

1 Overview

An Assembly program is made up of a sequence of instructions, each of which gets executed by the CPU. But why bother learning Assembly when we have C for embedded programming?

The main reason nowadays is that **Assembly** code is usually used as a **tiny** piece of a (larger) **C program** to do something **very fast and specialised**. And in this course, it also helps to give you an understanding of **low-level operations and register manipulation**. And after writing your own Assembly code, you should have a greater appreciation for all the simplicity inherent in C programming.

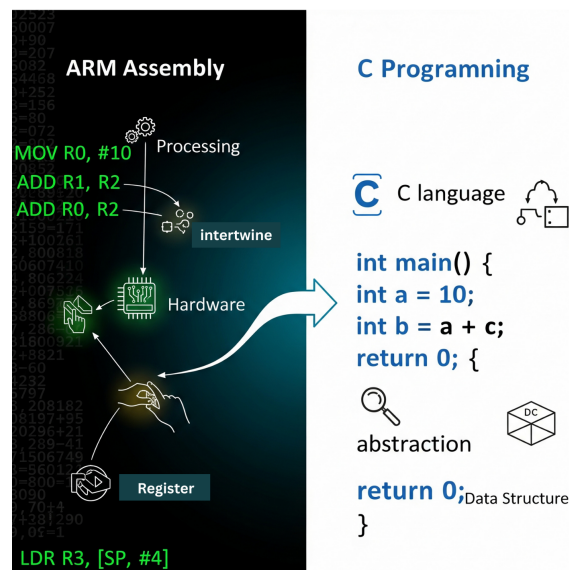


Figure 1: Welcome to Assembly

2 Outcomes and Knowledge Areas

In this practical, you will be writing Assembly code to **interface** with your **microcontroller** board to perform **basic operations**. After this practical, you should know how to manipulate the **LEDs** on your STM board and use the **pushbuttons** to change their sequencing — all of which is done in Assembly.

You will learn about the following aspects:

- **Interfacing** with the STM **board** using **Assembly**
- Using (and varying) **time delay loops** in Assembly
- Displaying **LED patterns** using Assembly
- **Reading** from **pushbuttons** using Assembly

3 Deliverables

For this practical, you must:

- Develop the code required to meet all objectives specified in the Tasks section
- Push your completed code to a shared repository on GitHub
- Demonstrate your working implementation to a tutor in the lab (**explicitly showing the use of Assembly code instead of C**). You will be allowed to conduct your demo during any lab session before the practical submission deadline.
- Write a short report documenting your **assembly.s** code, GitHub repo link, and a brief description of the implementation of your solutions. This must be in PDF format and submitted on Gradescope with the naming convention(**If you do not adhere to the naming convention, there will be a penalty**):

EEE3096S 2025 Practical 1 Hand-in STDNUM001 STDNUM002.pdf

Check the Appendix for Report Structure 6.

- Your practical mark will be based both on your demo to the tutor (i.e., completing the below tasks correctly) as well as your short report. Both you and your partner will receive the same mark.

Note: Your code (and GitHub link) should be copy-pasted into your short report so that the text is fully highlightable and searchable;**do NOT submit screenshots of your code (or repository link) or you will be penalised.**

4 Getting Started

The procedure is as follows:

1. Clone or download the Git repository(The practical folder of interest is Practical2:

```
git clone https://github.com/EEE3096S-UCT/EEE3096S-2025.git
```
2. • Open **STM32CubeIDE**, then navigate through the menus:

File → Import → Existing Code as Makefile Project

- Click **Next**.
- In the dialog, click **Browse...** and select your project folder.
- Under **Toolchain**, choose **MCU ARM GCC**.
- Click **Finish**.

Note: This IDE provides a GUI to set up clocks and peripherals (GPIO, UART, SPI, etc.) and then automatically generates the code required to enable them in the main.c file. The setup for this is stored in an .ioc file, which we have provided in the project folder if you would ever like to see how the pins are configured.**However, it is crucial that you do NOT make/save any changes to this .ioc file as it would re-generate the code in your main.c file and may delete code that you have added.**

3. In the IDE, navigate and open the **assembly.s** file under the Core/src folder, and then complete the Tasks below. **You will not be editing main.c in this practical.**

Note: All code that you need to write/add in the assembly.s file is marked with a "TODO" comment; **do not edit any part of the other code that is provided.**

5 Tasks

Complete the following tasks using the assembly.s file in STM32CubeIDE, and then demonstrate the working execution of each task to a tutor:

1. By default, the LEDs should increment by 1 every 0.7 seconds (with the count starting from 0).
2. While SW0 is being held down, the LEDs should change to increment by 2 every 0.7 seconds.
3. While SW1 is being held down, the increment timing should change to every 0.3 seconds.
4. While SW2 is being held down, the LED pattern should be set to 0xAA. Naturally, the pattern should stay at 0xAA until SW2 is released, at which point it will continue counting normally from there.
5. While SW3 is being held down, the pattern should freeze, and then resume counting only when SW3 is released.

Lula: SW0, SW1
Phila: SW2, SW3

Note: Only one of SW2 or SW3 will be held down at one time, but SW0 and SW1 may be held at the same time.

6 Helpful Assembly Tips

- Spending a few minutes beforehand and planning your code path (with a flowchart or some such) can make your coding a lot more efficient.
- Check the Course Notes for how to implement a basic delay.
- Use labels, conditions, and branches; you could always cut down on them later when optimising your code once you have a working solution.
- Do not edit the provided code base — especially the constants at the end of the code and frontmatter. These have been marked “DO NOT EDIT” for good reason!
- You only need to edit assembly.s in this practical; the .ioc file and .c files can be ignored and left as they are.

Extra resource on STM32 ARM Assembly programming can be found here:

1. [STM32 Programming Manual](#)
2. [Arm Assembler Tutorial](#)

Appendix: Report Structure

Section	Description
Introduction	Briefly introduce the aim and objectives of the practical and summarise your work briefly.
Methodology	Detail the steps undertaken and methods used to achieve the practical task in a logical and coherent manner.
Results and Discussion	Present the results and discuss their significance.
Conclusion	Summarise the work you did for the practical and any improvements you could make to your implementation.
AI Clause	In one paragraph discuss how you used LLMs while working on the practical and if you did find them useful in an embedded systems programming context.
Appendix	Attach your <i>assembly.s</i>